THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. AC means 1

(a) always current **(b)** amplified current

(c) alternating current **(d)** ampere current

2. Which of these is a magnetic substance? 1

(a) Wood **(b)** Plastic **(c)** Iron **(d)** Glass

3. An electromagnet uses a core of 1

(a) plastic **(b)** copper **(c)** aluminium **(d)** soft iron

4. DC means 1

(a) dynamic current **(b)** direct current **(c)** double current **(d)** driving current

5. A magnet attracts 1

(a) iron nails **(b)** glass beads **(c)** wooden rods **(d)** plastic spoons

6. An electromagnet is a 1

(a) insulator **(b)** temporary magnet

(c) permanent magnet **(d)** weak natural magnet

7. Electric current from a battery is 1

(a) AC **(b)** DC **(c)** both **(d)** none

8. Outside a magnet, field lines emerge from the 1

(a) north pole **(b)** middle **(c)** south pole **(d)** both poles

9. Electromagnets are used in 1

(a) mirrors **(b)** torches only **(c)** glass **(d)** electric bells

10. The household supply in India is 1

(a) 110 V, 60 Hz AC **(b)** 220 V, 50 Hz AC **(c)** 12 V DC **(d)** 220 V DC

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define alternating current. 2

12. Who discovered that current produces a magnetic field? 2

13. Name the rule that gives the direction of force on a current-carrying conductor. 2

14. What is the use of the earth wire? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. List three differences between a permanent magnet and an electromagnet. 3

16. State the frequency of AC in India and explain what it means. 3

17. What is the use of slip rings and brushes in a generator? 3

OR

State the factors affecting the strength of an electromagnet. 3

18. List two advantages of AC. 3

OR

What do you understand by the term 'induced current'? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain Fleming's left-hand rule and state the factors on which the force on a conductor depends. 5

OR

Why is earthing needed? Explain with the help of a metal body appliance. 5

20. Describe Faraday's experiment of electromagnetic induction using a coil, magnet and galvanometer. 5

OR

Compare a generator and an electromagnet in terms of principle and energy change. 5

END OF QUESTION PAPER — Total: 40 Marks

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